



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

SpaceX Falcon 9 First-
Stage Landing
Prediction
29 September 2026



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Executive Summary

Data: IBM course snapshots of SpaceX launches and launch geography.

EDA: 90 Falcon 9 flights; 60 successful landings (66.7%).

Models: four classifiers each scored 83.3% on 18 test records.

Decision: no unique winner on the held-out test set.

Introduction

Can launch attributes help predict whether the first stage lands?

Reusable first stages can reduce launch cost. We examine launch site, payload, orbit, booster history and flight sequence.

The analyses use historical course datasets, not current launch records.

Section 1

Methodology

Methodology

Collect: archived SpaceX API snapshot and launch table.

Prepare: select Falcon 9, handle missing payload mass, encode outcome.

Explore: charts, SQL and launch geography.

Predict: tune and compare four classification models.

Data Collection

IBM course snapshots supply the launch and geography records.

API workflow: launches → rocket, payload, launchpad and core details → tabular records.

Wikipedia workflow: archived launch table → HTML parsing → launch rows.

The downstream analyses use the provided fixed course snapshots for reproducibility.

Data Collection – SpaceX API

Archived API launch JSON → select launch fields → enrich rocket, payload, launchpad and core attributes → Falcon 9 records.

The live SpaceX API returned an SSL 525 error during this run. The analysis therefore uses the IBM course snapshot.

Source notebook: SpaceX API collection lab.

Launch JSON



Selected fields



Falcon 9 table

Data Collection - Scraping

Archived Wikipedia page →
identify launch tables → parse
date, booster, site, payload, orbit
and landing status → tabular
file.

The notebook contains the
completed parsing code. Live page
retrieval was not verified in this
run.

Source notebook: web scraping
lab.

Archived page
↓
HTML tables
↓
Launch records

Data Wrangling

- Select Falcon 9 records and number flights sequentially.
- Fill missing payload mass with the observed mean.
- Define Class = 1 for successful first-stage landings; otherwise 0.
- One-hot encode orbit, site, landing pad and serial; cast features to float.
- Final model matrix: 90 rows $\times$ 83 columns.

EDA with Data Visualization

Scatter plots compare outcomes with flight number, payload, site and orbit.

Bar chart compares landing success by orbit.

- A yearly line chart shows how the sample success rate changed.
- All six charts were executed in the EDA notebook.

EDA with SQL

SQL tasks identify sites and CCAFS records, aggregate payload mass, date the first ground landing, and rank landing outcomes.

Queries ran against the fixed 101-row launch table.

- Results appear on slides 24–33.

Build an Interactive Map with Folium

Geographic dataset contains 56 launch records and four site labels.

Site coordinates, outcome markers and proximity points explain where launches occurred.

Map visuals in this deck are static views made from the same geography table.

Build a Dashboard with Plotly Dash

The dashboard views summarize success counts by site, success ratio at the leading site, and payload versus outcome.

Charts on slides 39–41 were reconstructed from the 90-row analysis dataset. No live Dash application was supplied.

Predictive Analysis (Classification)

Split 90 records into 72 training and 18 test rows.

Standardize numeric predictors and tune logistic regression, SVM, decision tree and KNN with 10-fold cross-validation.

Compare accuracy on the same held-out test set; inspect a confusion matrix.

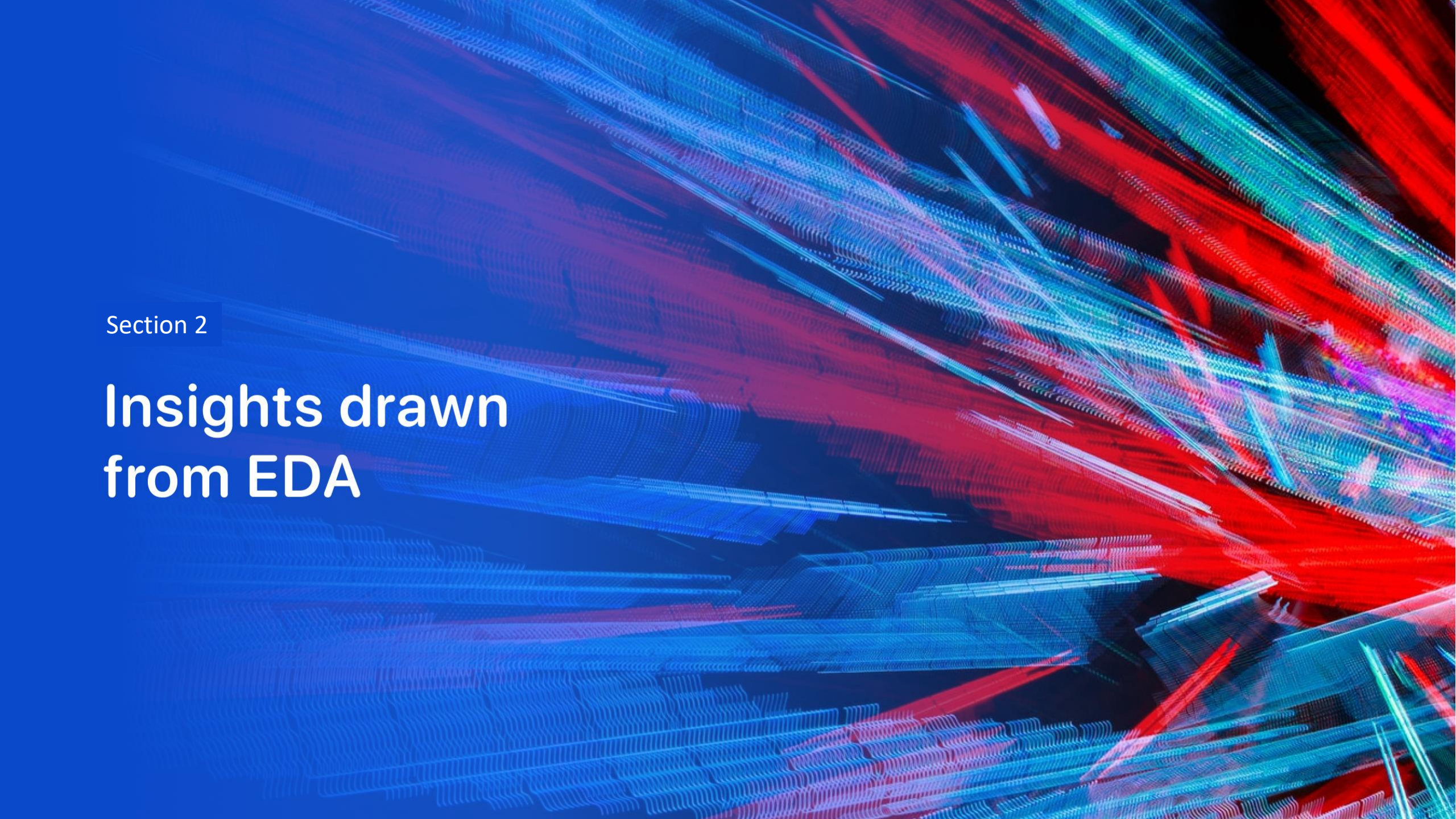
Results

EDA: site and orbit patterns, plus a rising historical trend.

SQL: first successful ground landing on 22 December 2015.

Geography: four labels span Florida and California.

Prediction: every tested classifier achieved 15/18 correct.



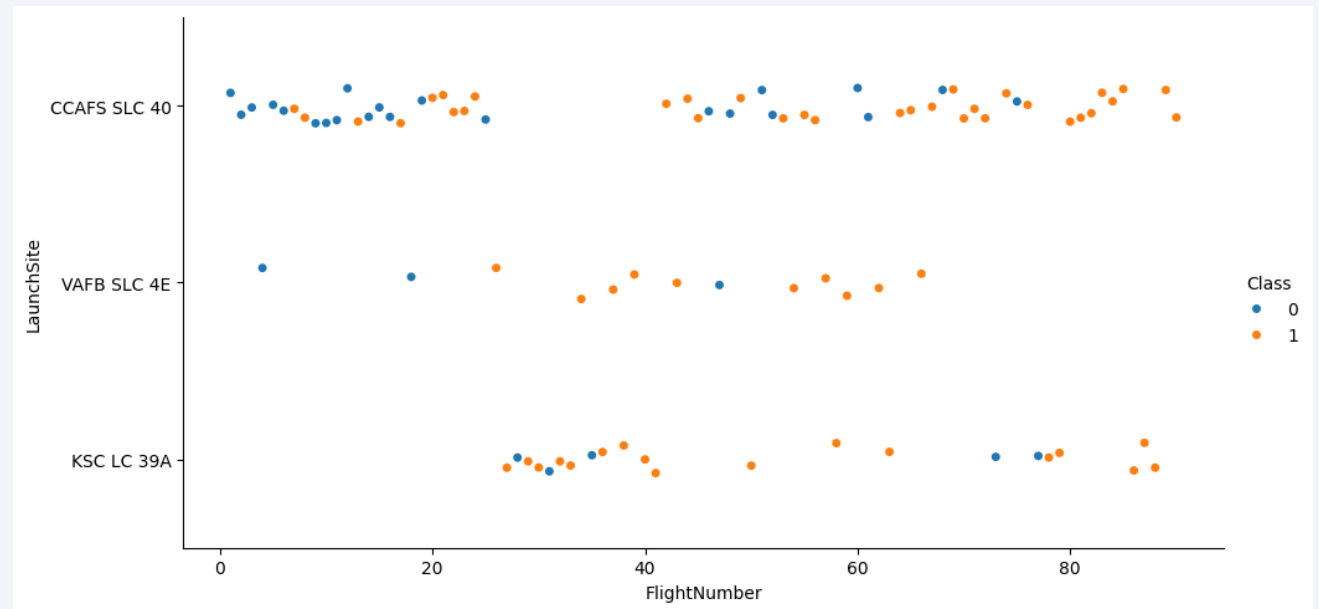
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Later flight numbers show more green outcomes across the sites.

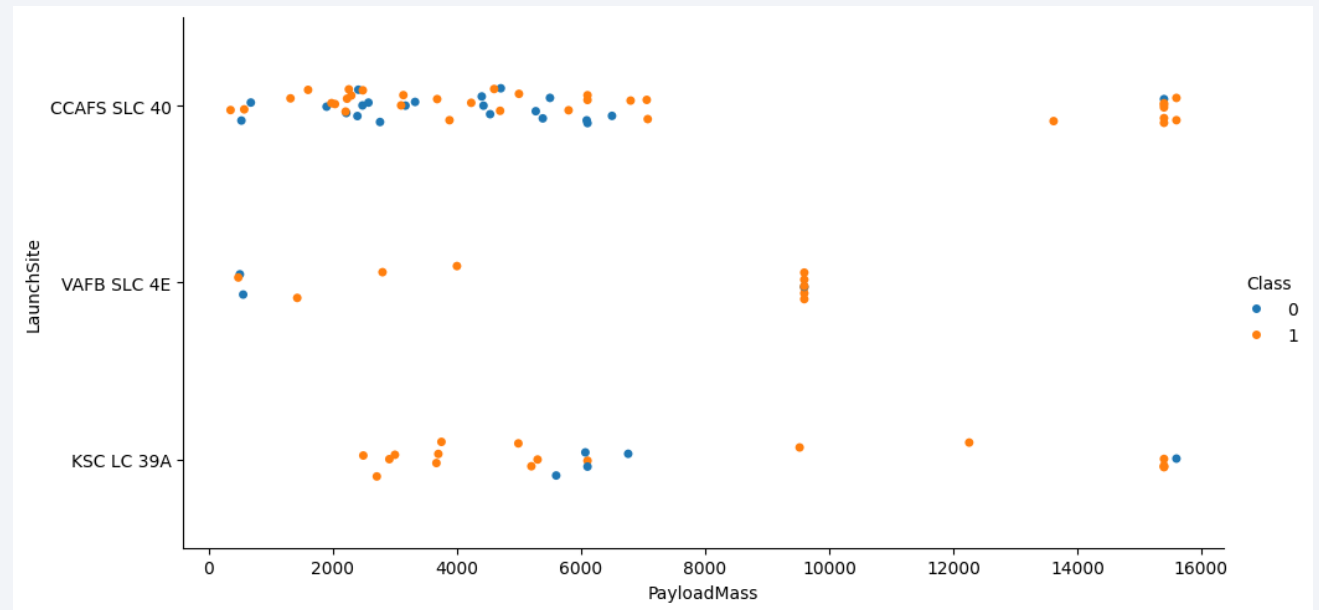
CCAFS SLC 40 contains the most observations (55 of 90).



Payload vs. Launch Site

Payload mass overlaps for successful and other outcomes.

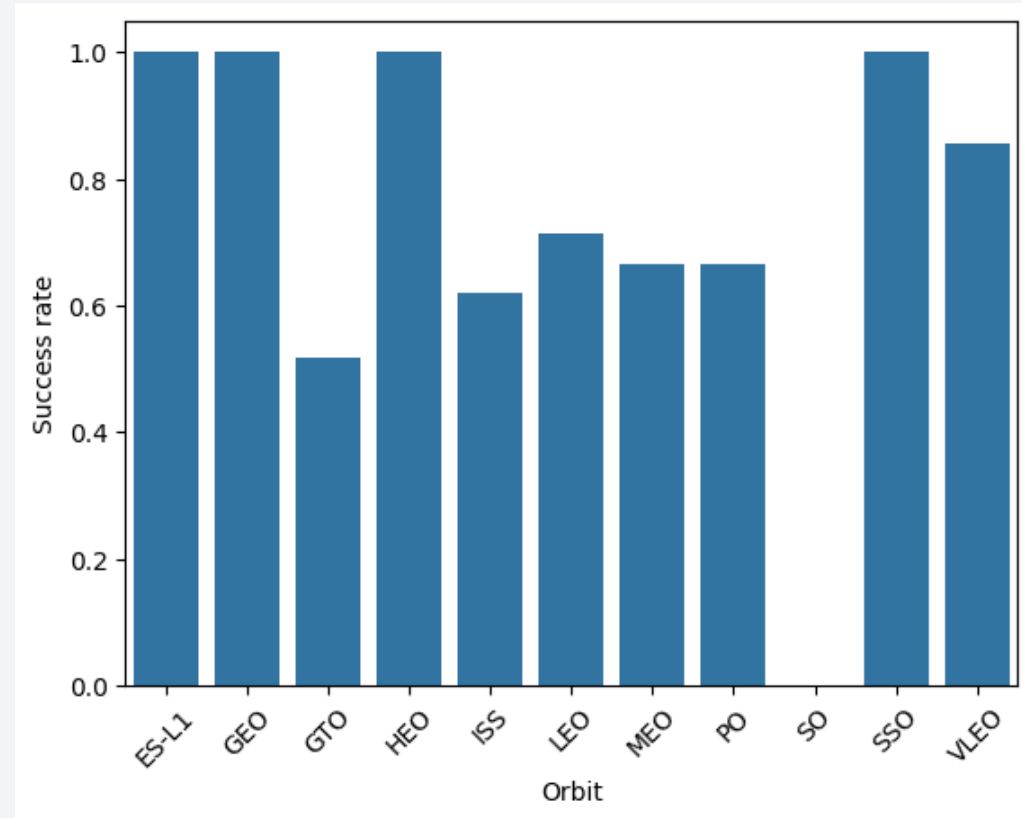
This sample does not show a single payload cutoff that separates them.



Success Rate vs. Orbit Type

Success varies by orbit in the historical sample.

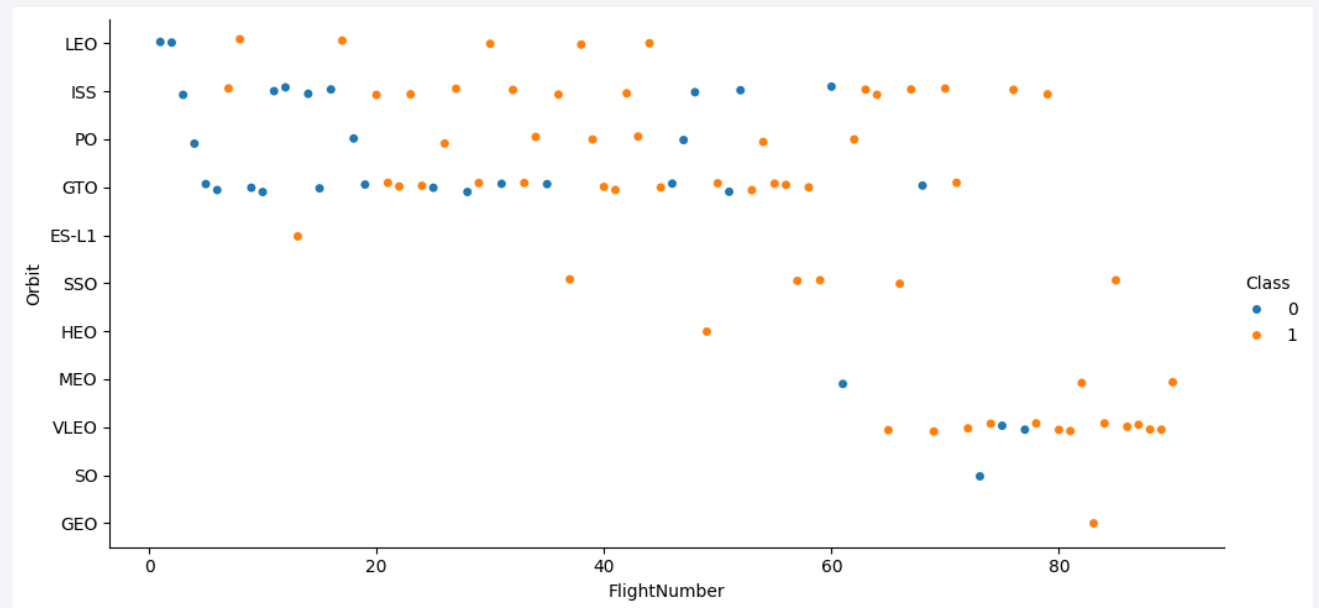
Interpret small orbit categories cautiously because a high percentage can reflect few launches.



Flight Number vs. Orbit Type

LEO later flights show more successful landings.

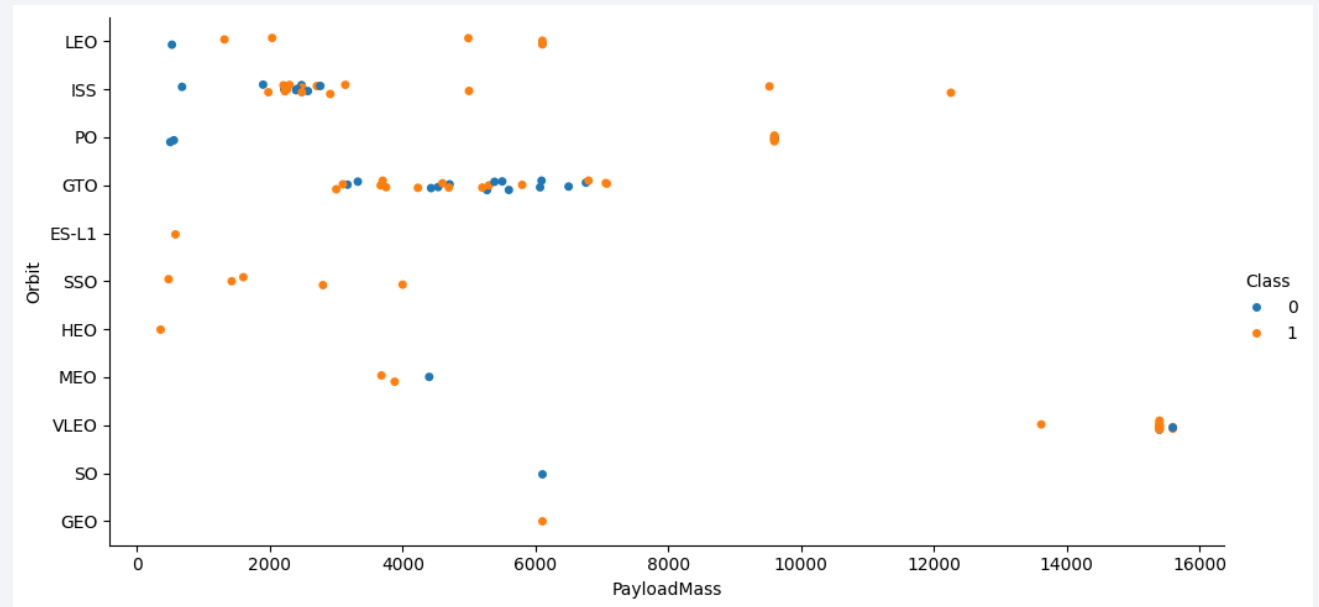
GTO records show both outcomes over the observed flight sequence.



Payload vs. Orbit Type

The relationship between payload and landing outcome differs across orbit categories.

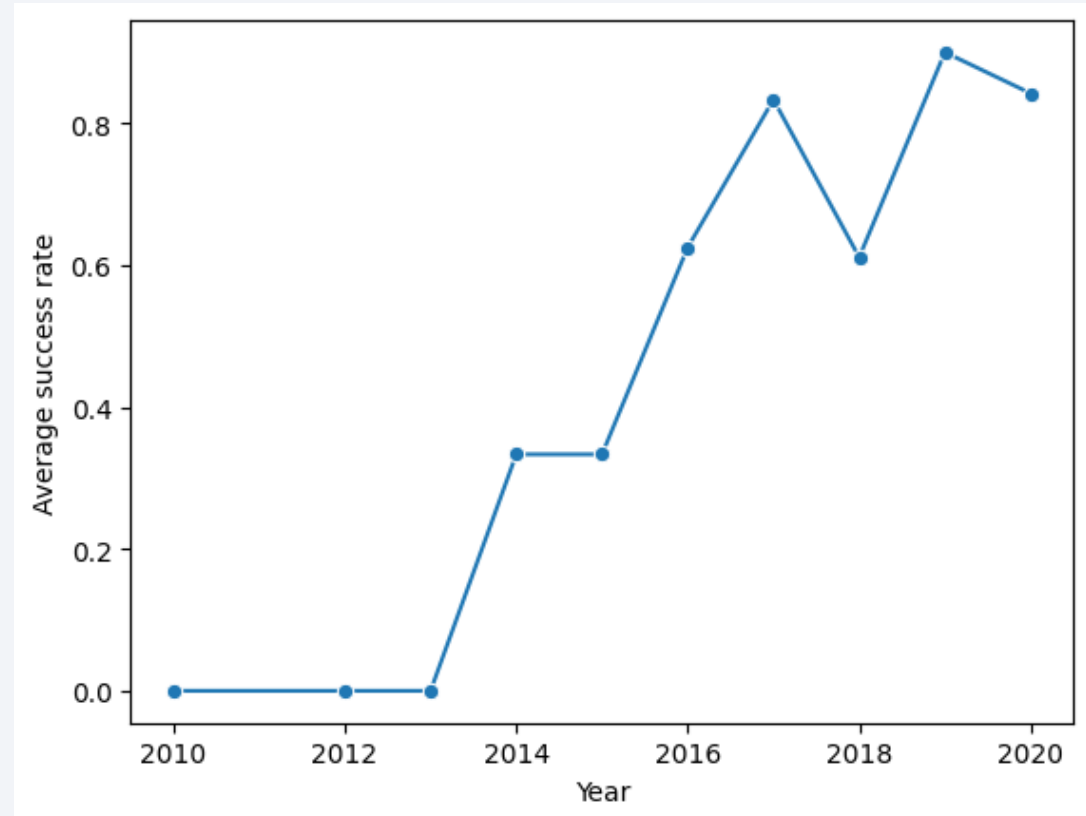
GTO has overlapping successes and other outcomes.



Launch Success Yearly Trend

The yearly mean landing success rate rises across the historical sample, with year-to-year variation.

This is a descriptive trend, not a forecast.



All Launch Site Names

Four distinct site labels: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A and VAFB SLC-4E.

The two CCAFS names appear separately in the SQL source.

Launch Site Names Begin with 'CCA'

The first five CCA-prefixed records start in 2010 at CCAFS LC-40.

The SQL query filters `launch_site` LIKE “CCA%” and limits the output to five rows.

Total Payload Mass

NASA (CRS) missions carried 45,596 kg of payload in the 101-row SQL dataset.

Average Payload Mass by F9 v1.1

F9 v1.1 boosters carried an average payload of 2,534.67 kg in the SQL dataset.

First Successful Ground Landing Date

The first successful ground-pad landing in the SQL records occurred on 22 December 2015.

Successful Drone Ship Landing with Payload between 4000 and 6000

Successful drone-ship landings with 4,000–6,000 kg payload include B1022, B1026, B1021.2 and B1031.2.

Total Number of Successful and Failure Mission Outcomes

Mission outcomes in the source: 98 “Success”, one “Success ” with trailing space, one “Success (payload status unclear)” and one “Failure (in flight)”.

The strings are shown separately because the SQL source preserves their exact spelling.

Boosters Carried Maximum Payload

Several F9 Block 5 booster flights share the maximum payload in the SQL table.

The query returns every booster whose payload equals the maximum, rather than choosing one arbitrarily.

2015 Launch Records

Two 2015 drone-ship landing failures appear in the filtered records: F9 v1.1 B1012 and B1015, both from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

From 2010-06-04 through 2017-03-20, “No attempt” ranks first (10), followed by success and failure on drone ship (5 each).

Successful ground-pad landings appear 3 times.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue rectangle on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible, separating the dark surface from the deep blue of the atmosphere and the blackness of space.

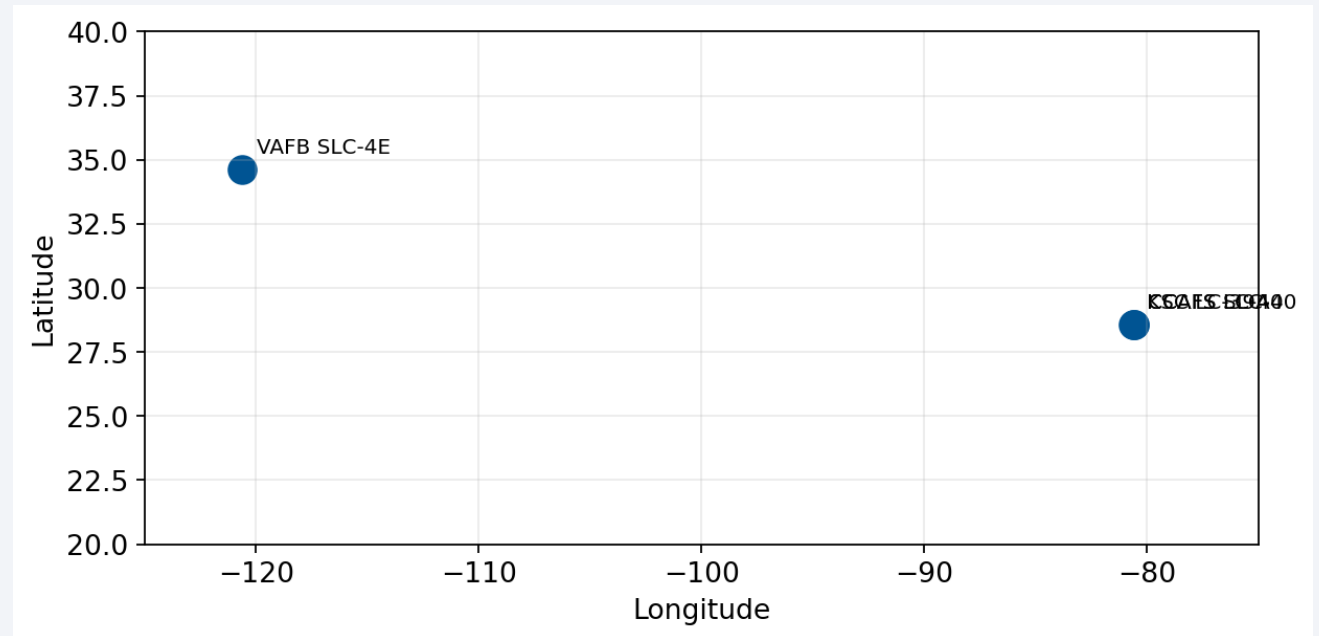
Section 3

Launch Sites Proximities Analysis

Launch Site Locations

Four launch-site labels occur in the geography table.

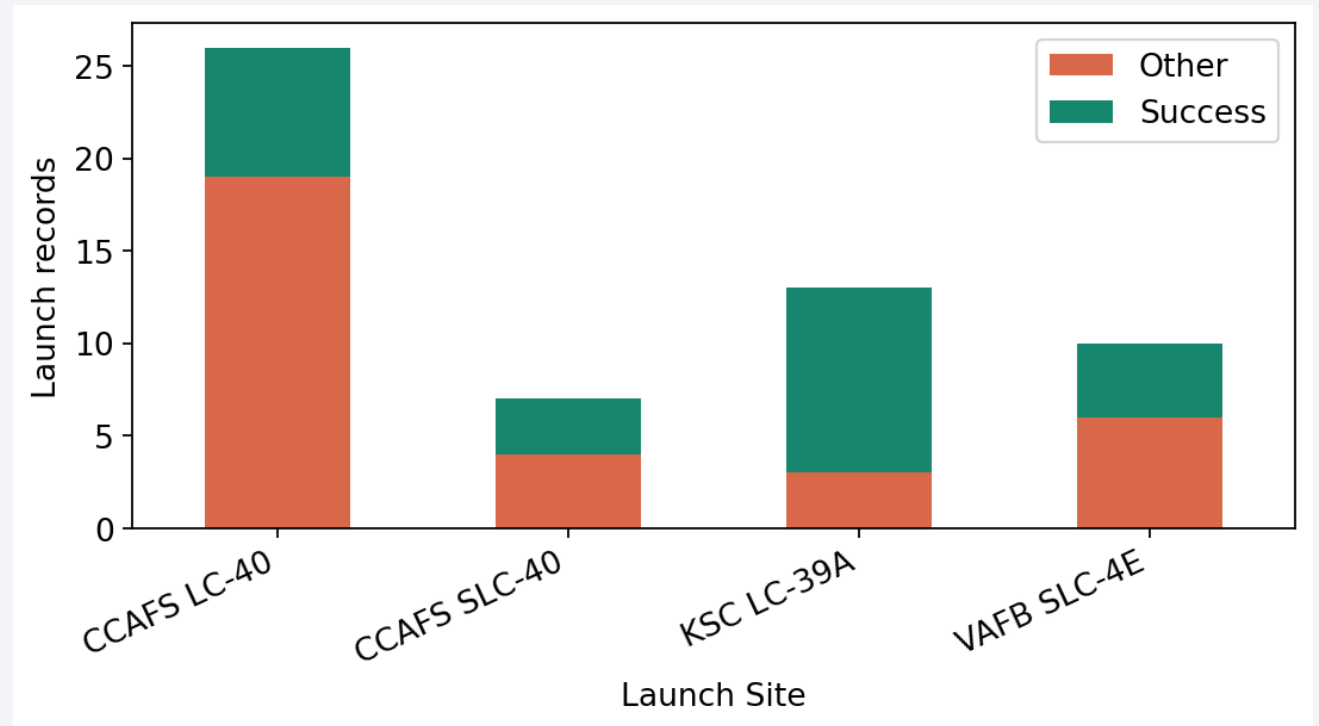
Three are in Florida and VAFB SLC-4E is in California.



Landing Outcomes by Launch Site

The 56-row geography snapshot marks each launch as successful or other.

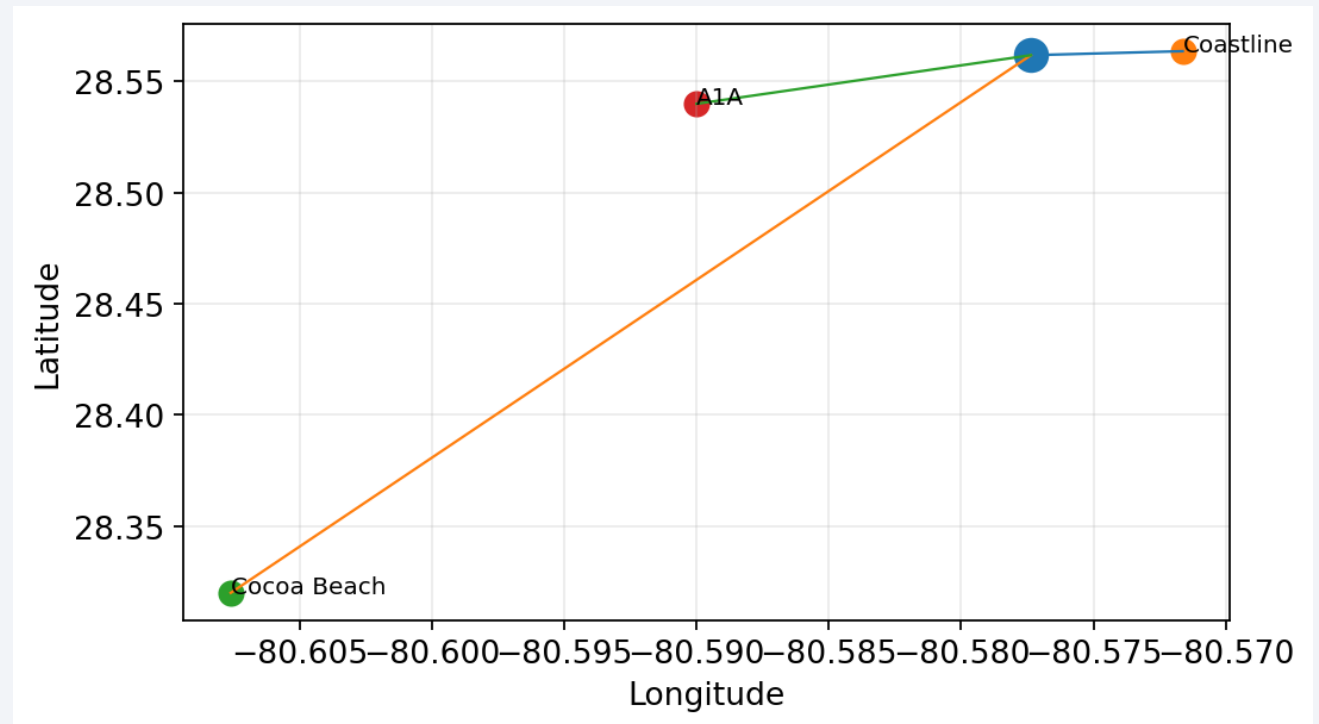
Stacked counts show the outcome distribution at each site.

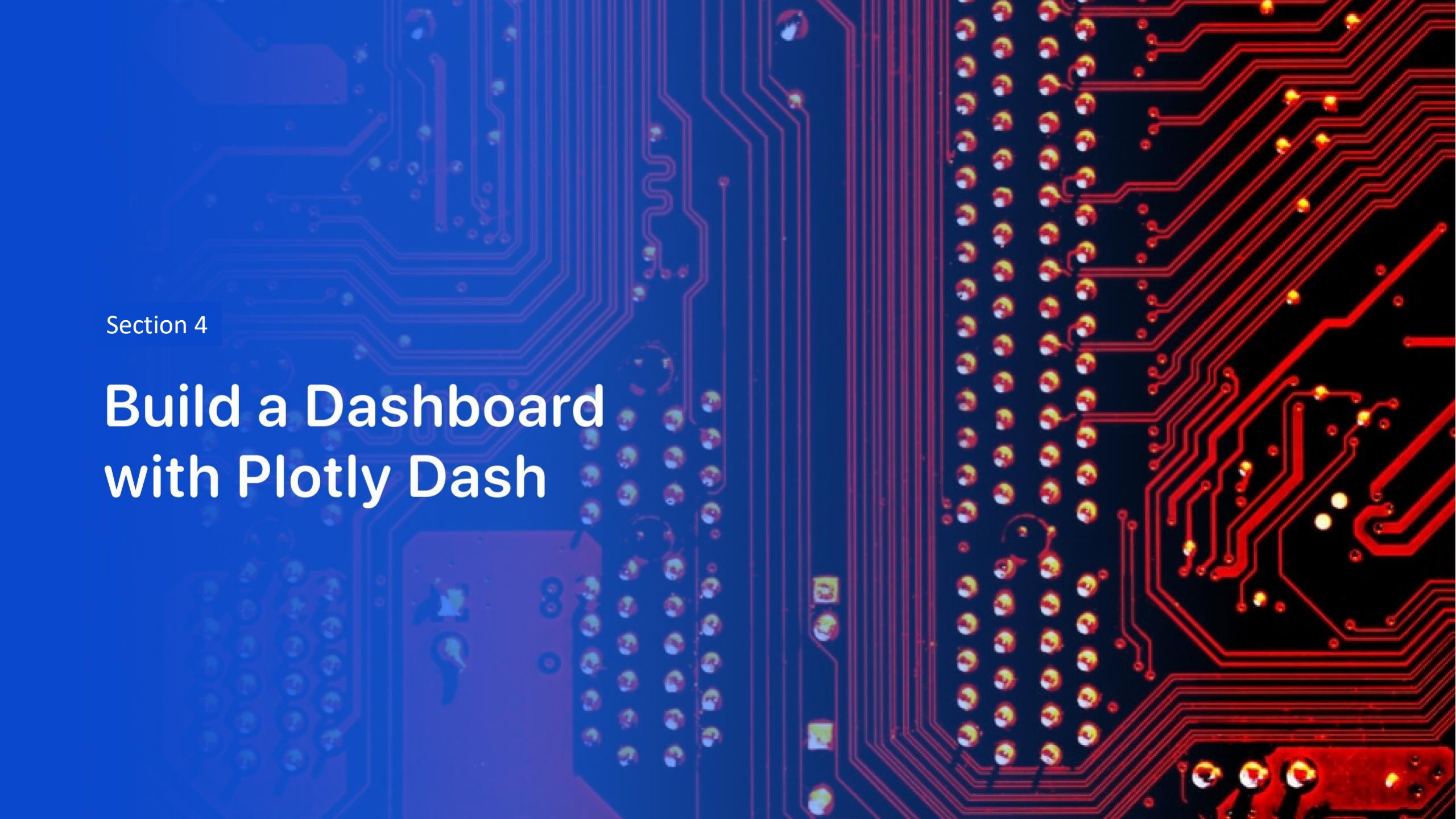


Launch Site Proximity

Example from CCAFS SLC-40: the coastline point is about 0.62 km away using the haversine formula.

Other marked points are illustrative nearby landmarks; distances depend on the selected coordinates.





Section 4

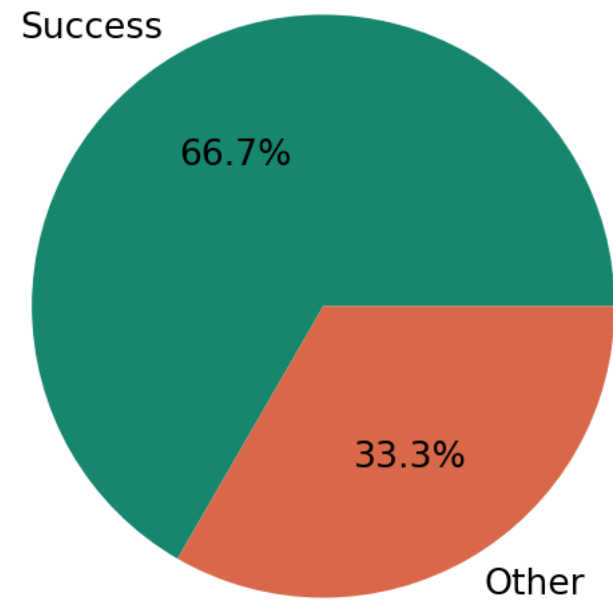
Build a Dashboard with Plotly Dash

All-Site Landing Outcomes

Across the 90-row analysis dataset, 60 first stages landed successfully and 30 had other outcomes.

Overall sample success rate: 66.7%.

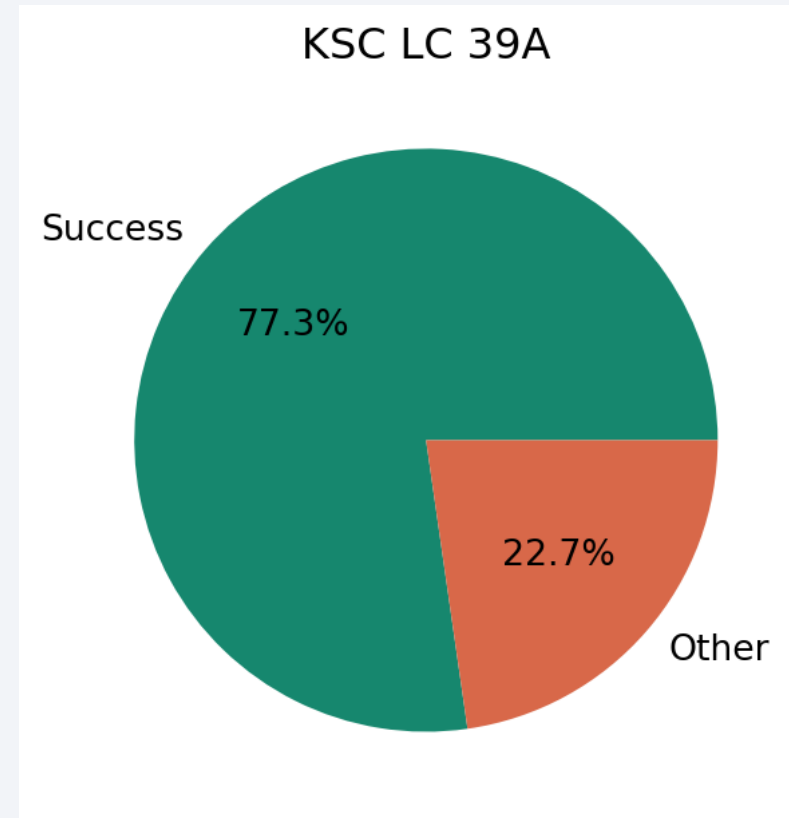
All sites, first-stage landing outcome



Highest Site Success Share

KSC LC 39A has the highest observed success share: 17 of 22 launches, or 77.3%.

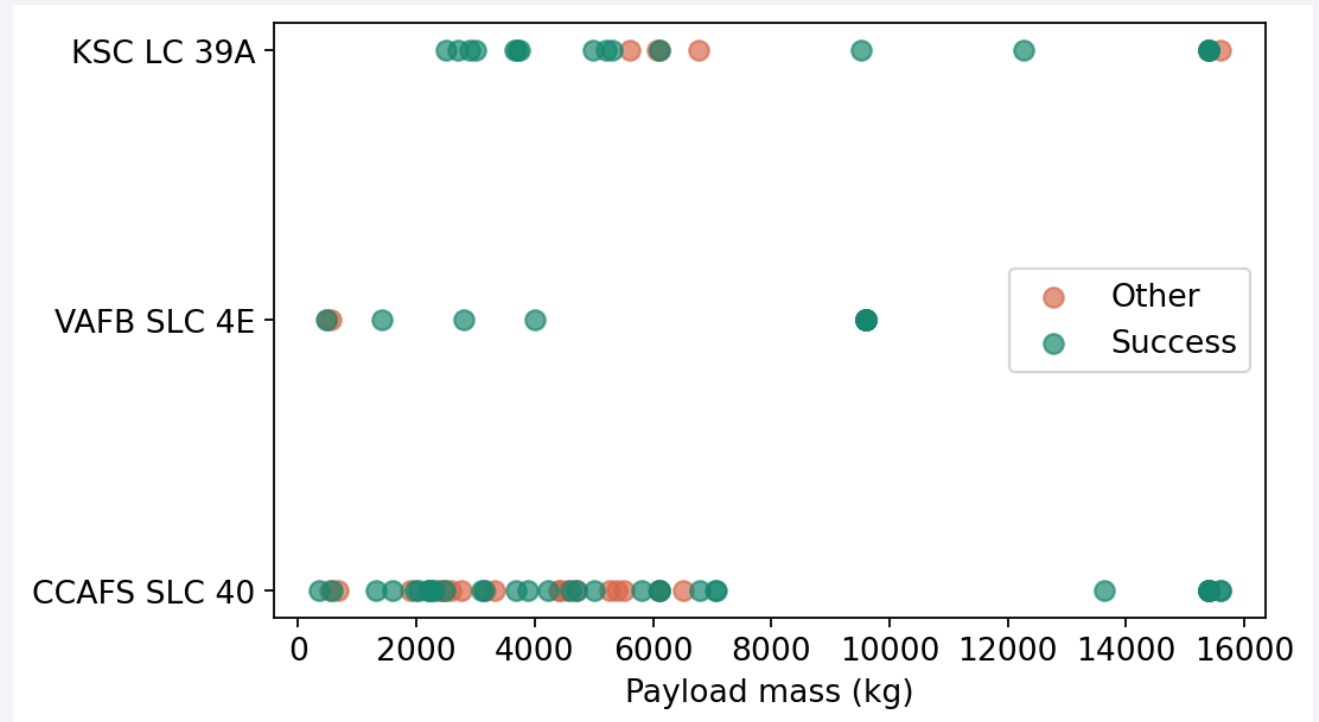
VAFB SLC 4E is close at 10 of 13 (76.9%).



Payload and Landing Outcome

Both outcome classes occur across a broad payload range.

Filtering by payload can change counts, but the static sample does not yield a clean threshold.



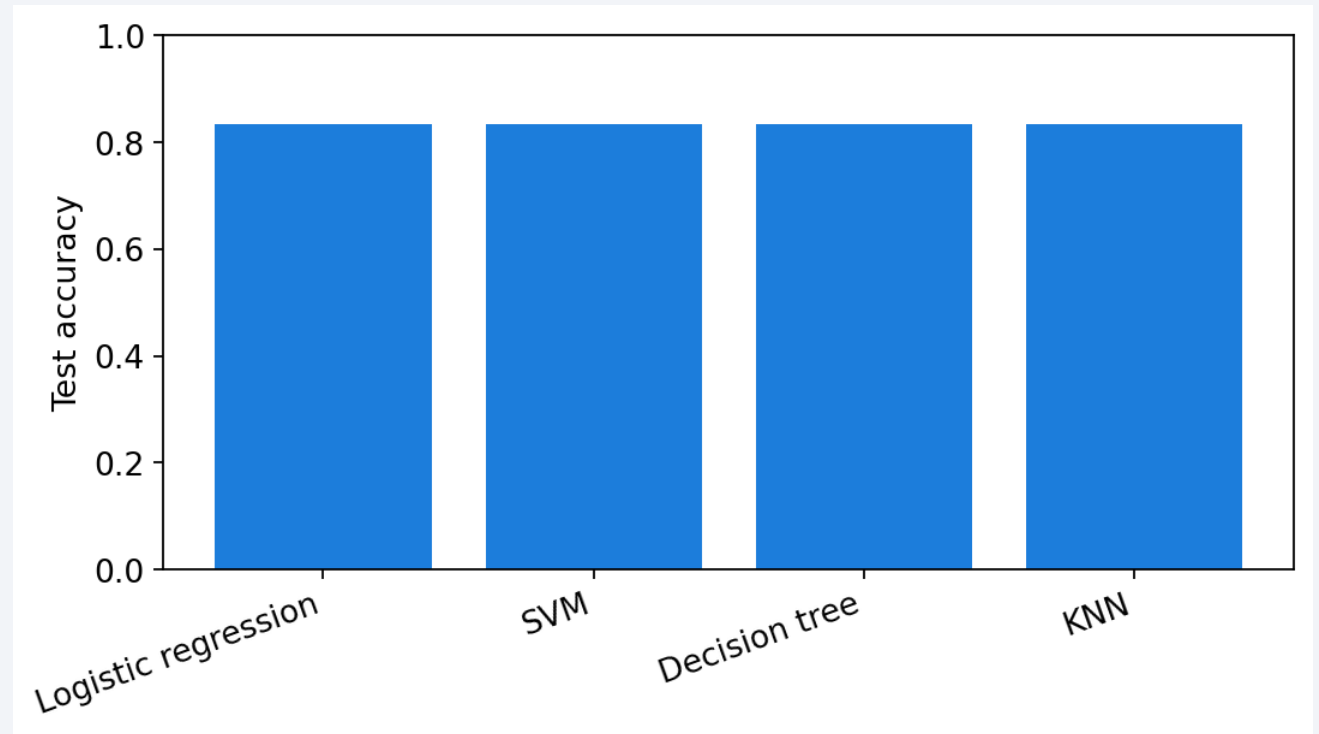
Section 5

Predictive Analysis (Classification)

Classification Accuracy

Held-out accuracy is 0.833 for logistic regression, SVM, decision tree and KNN (15/18 correct).

Decision tree has the highest cross-validation mean (0.877), but there is no unique test-set winner.

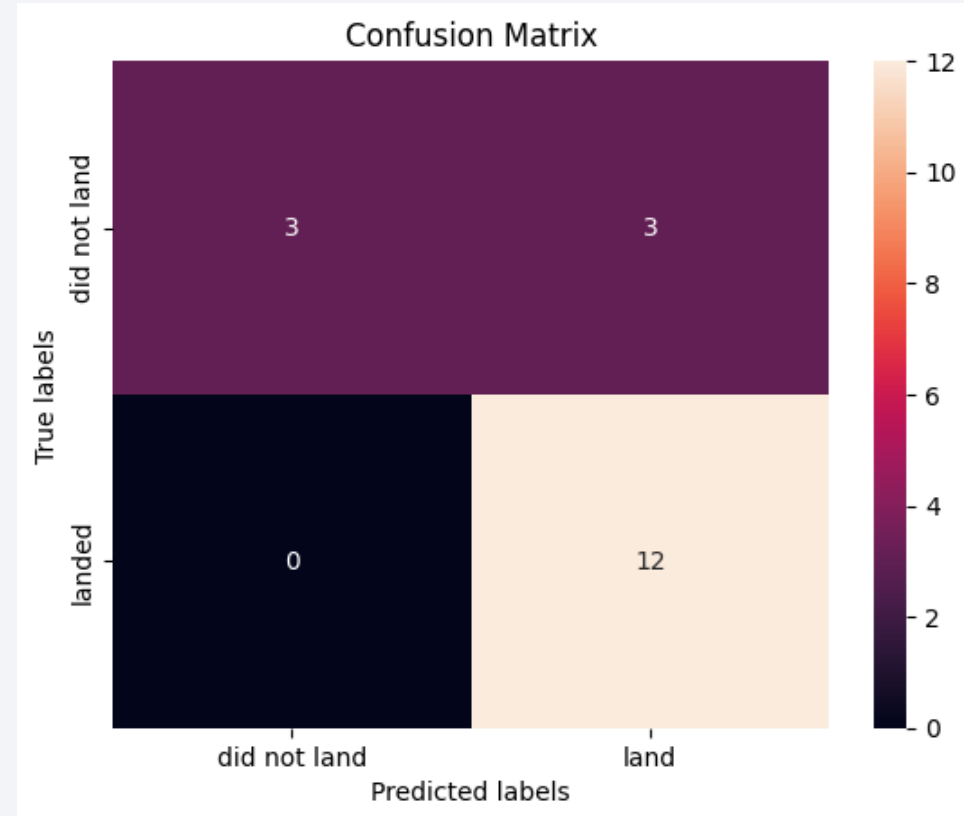


Confusion Matrix

Logistic regression confusion matrix on 18 test flights.

15 predictions are correct and 3 are incorrect.

A tie in test accuracy means this is one representative model, not a proven winner.



Conclusions

The historical 90-flight sample has a 66.7% successful landing rate.

Flight sequence and launch context show useful descriptive patterns.

All four tuned classifiers tie at 83.3% test accuracy on 18 records.

A larger independent test set is needed to distinguish models reliably.

Appendix

Data sources: IBM SpaceX course snapshots, dataset_part_2.csv and spacex_launch_geo.csv.

Executed notebooks: SQL, EDA visualization and machine learning.

Methods: pandas, SQLite, seaborn, scikit-learn.

Live API retrieval and the interactive Dash application were not validated in this run.

Thank you!

